Type 1: Research Project

Structural Descriptors for Metal–Organic Frameworks:
Multifractal Analysis of Framework Connectivity for Simulation-Free Screening

23CSE498 – Project Phase 2 (Panel Review 1) | Team Number: [XX]

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0. Problem Context — Why This Project Exists

The gap between the number of candidate materials and our ability to evaluate them

The materials-science problem

- **Metal–organic frameworks (MOFs)** are crystalline solids built by joining metal clusters (*nodes*) to organic molecules (*linkers*).
- They are extremely porous — internal surface areas above 3,000 m^2/g — making them candidates for gas storage, carbon capture and separation.
- **Hundreds of thousands** of MOFs are known or hypothesised.

The computational bottleneck

- Evaluating one MOF by molecular simulation (GCMC) takes **hours**.
- Exhaustive screening of the design space is **not feasible**.

Research Question

Can a **purely structural, simulation-free descriptor** — computed from a framework's *connectivity* in seconds — group MOFs into meaningful families, so a new candidate can be triaged **without** expensive calculation?

Value Proposition

Not a *replacement* for simulation — a **filter** that decides **what to simulate**. Reducing 10^5 candidates to 10^2 is the contribution.

1. Methodology & Solution Design

Rubric: Workflow, Architecture, Algorithms/Mathematical Model & Evaluation Metrics (5 Marks)

Theoretical foundation. A MOF is a **periodic graph**. Collapse each building block to a vertex and each inter-block bond to an edge; the chemistry reduces to pure connectivity, which the properties follow.

Core algorithms

- **Decomposition:** MOFid metal-oxo algorithm [1] — PBC-aware bond perception, connected-component block extraction.
- **Graph construction:** *labelled quotient graph* — edges carry a lattice translation t, so the infinite crystal is represented exactly.
- **Descriptor:** influential-node multifractal analysis (iNMFA) [2].

Dataset. 77 frameworks queried live from MOFX-DB [7], with published PLD/LCD pore diameters. Supercells of 3,400–6,100 atoms.

CIF
cell + fractional co-ords (*no bonds stored*)

↓

Bond perception
min-image over 27 cell images, covalent-radius cutoff [10]

↓

Block decomposition
metal-oxo split → nodes, linkers

↓

Labelled quotient graph
edge = $(A, B, \boldsymbol{t})$;
supercell expansion

↓

iNMFA
box covering →
$\mathcal{P}_q(r) \rightarrow \tau(q) \rightarrow f(\alpha)$

↓

Descriptor
$\Delta\alpha$, α_0, $A \Rightarrow$ the band

Validation metric: published **coordination numbers** — an independent ground truth the graph must reproduce.

1b. Mathematical Model

The descriptor, defined

Step 1 — the measure. For each influential node i (top 10% by degree) and box radius r, let $M_i(r)$ be the number of vertices within r steps:

$$p_i(r) = M_i(r)/N$$

Step 2 — the partition function, reweighted by a distortion factor q:

$$\mathcal{P}_q(r) = \sum_{i \in I} p_i(r)^{\,q}$$

Step 3 — the mass exponent.

$$\boxed{\tau(q) = \frac{\ln \mathcal{P}_q(r)}{\ln(r/r_N)}}$$

Step 4 — Legendre transform.

$$\alpha(q) = \frac{d\tau}{dq}, \quad f(\alpha) = q\,\alpha(q) - \tau(q)$$

Interpreting the spectrum

Quantity	Physical meaning
α	local growth rate around one block
$f(\alpha)$	how much of the framework shares it
α_0	the *typical* dimension (apex)
$\Delta\alpha$	heterogeneity of the pore network
A	asymmetry — dense-vs-open dominance

Structural constraint used as a correctness test

$f(\alpha)$ must be an inverted parabola peaking at $q = 0$, where $f(\alpha_0) = D_0$. Any implementation not producing this shape is wrong — we used it as a test, and it caught a real defect (Slide 3).

Evaluation metrics: coordination-number agreement with published crystallography; spectrum shape (% peaking at $q = 0$); seed reproducibility; partial correlation against pore diameter.

2. Implementation Progress & Module Functionality

Rubric: Core Modules Implemented, Functional & Integrated (8 Marks)

Module / Component	Current Implementation State	Status
Data ingestion (`code_01-02`)	CIF parsing, symmetry expansion, PBC bond perception over 27 cell images	Functional
Decomposition (`code_03-04`)	`is_metal()` classification; metal-oxo, single-node, all-node splits	Functional
Topology export (`code_05-07`)	Centroid simplification, Systre `.cgd` writer, MOFid/MOFkey assembly	Functional
Network analysis (`code_09`)	Labelled quotient graph, degree/eigenvector/ betweenness centrality, Laplacian	Functional
Spectrum (`code_10-11`)	iNMFA (box-covering) and NMFA (box-growing); τ, α, $f(\alpha)$, $\Delta\alpha$, A	Functional
77-framework study	Live MOFX-DB query, spectra for all 77, band + confound analysis	Functional
Interactive web platform	18-section site; full pipeline runs live in-browser on any uploaded CIF	Integrated
Size-confound correction	Fit-window normalisation / size-matched comparison	In Progress
Experimental-set validation	Re-query CoRE MOF [9] for genuine metal diversity	Planned

How the ≈60% is arrived at (not asserted). Engineering ≈80%: 14 Python modules (2,681 lines), a cross-validated browser implementation (16 JS modules), an 18-section platform, 8 automated test suites, a 77-framework study running end to end. **Scientific validation ≈35%**: the descriptor is computed correctly, but its *physical meaning* is not yet established — the size confound is unresolved and the dataset is hypothetical and single-metal. **Weighted: ≈60%.** **Verification:** Python and JavaScript agree on every reported value ($\tau(0) = -2.6551$ on HKUST-1 in both).

Code: **github.com/Radha-Krishna-19/fyp**

2b. Results Obtained So Far

Every number computed from the live pipeline — nothing transcribed

Validation — the graph is correct

	Node c.n.	Published	Net
HKUST-1	4	$(4,3)$	`tbo`
MOF-5	6	$(6,2)$	`pcu`
ZIF-8	4	$(4,2)$	`sod`
UiO-66	**12**	$(12,2)$	`fcu`

All four match. A naive construction gives UiO-66 $=$ **1**.

Defects make influence measurable

In a perfect crystal every block is symmetry-equivalent $\Rightarrow$ only **2** distinct centrality values $\Rightarrow$ "top-k" is a **tie**. Remove one linker: $2 \rightarrow 4$ distinct values, and $\lambda_2 : 1.44 \rightarrow 1.19$ — the weakening, quantified.

The band — 77 frameworks

Spectra peaking at $q = 0$	**77 / 77**
$\Delta\alpha$ range	1.016 – 1.410
Interquartile band	**1.112 – 1.250**
Asymmetry A	-1.51 to -1.04 (all negative)

A tight band with a consistent asymmetry sign.

What the band does *not* yet mean

$\Delta\alpha$ vs pore diameter, raw	-0.497
controlling for graph size	$+0.101$
$\Delta\alpha$ vs graph size, ctrl. pore	$+0.621$
R^2: size alone / +pore	0.533 / 0.538

Pore size adds **+0.005**. $\Delta\alpha$ tracks **supercell size** — a computational parameter, not chemistry.

3. Technical Knowledge & Design Decisions

Rubric: Understanding of Methodology, Implementation & Design Choices (5 Marks)

Why a graph representation? MOF properties follow *connectivity*: gas moves through the pore network, and the pore network is the void left by the wiring. An **unweighted** graph deliberately discards element identity — what survives is **pure architecture**, which transfers across chemistries.

Key design decisions

- **Labelled quotient graph over a naive graph.** Without translation labels, UiO-66's 12-connected cluster reports as **1** — or degenerates into a star.
- **Supercell expansion.** A primitive cell has diameter $\sim$2, too small for a power-law fit. Expansion is *exact* because edges carry translations.
- **Box-growing (deterministic) validated against box-covering (stochastic).**
- **Defect simulation** to break symmetry — physically justified by [15].

Defect found by our own shape test

The spectra were not parabolic. Cause: $\tau(q)$ was extracted with a **free-intercept** least-squares slope, but [2] defines it as a **ratio** $\ln \mathcal{P}_q / \ln(r/r_N)$ — a fit **through the origin**. At $q = 0$: every term is $p_i^0 = 1$, so $\mathcal{P}_0 = |I|$ is **constant in** r. A flat line has slope $0 \Rightarrow \tau(0) = 0 \Rightarrow f(\alpha_0) = 0$ — **the peak is deleted**.

τ definition	$\tau(0)$	$f(\alpha_0)$	parabola
free intercept	0.0000	0.0000	no
paper (ratio)	-2.5669	$+2.5669$	**yes**

Trade-offs. `MIN_CELL_LENGTH`/`MAX_ATOMS` trade scaling range against $O(N^2)$ memory — and are the source of the size confound we identified. Influential-node selection is ambiguous in a perfect crystal (all degrees tie); we report the ambiguity rather than hide it.

3b. Deliverable — Interactive Analysis Platform

An 18-section technical site; the full pipeline executes in-browser

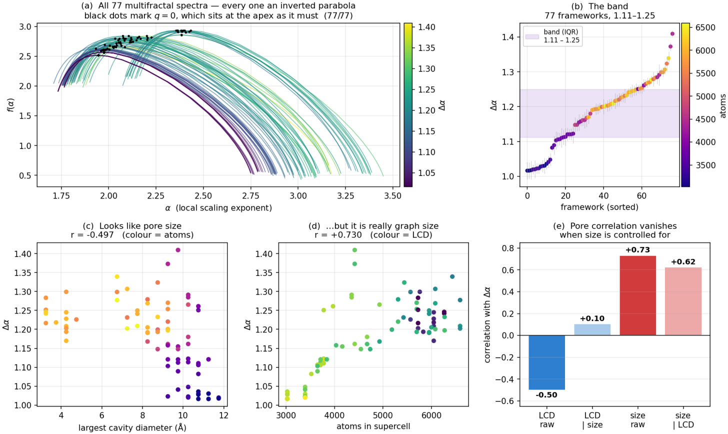
What it does

- Runs the **complete decomposition live** on any uploaded `.cif` — no server, no installation.
- Steps through all nine pipeline modules with the **real source code beside a live 3D model** of the structure at that stage.
- Computes the network analysis and the multifractal spectrum **in the browser**, cross-validated against the Python reference.

Built for a mixed audience

- A **graph-theory primer** (Section 10) defines every term — vertex, degree, quotient graph, Laplacian, λ_2, defect — *before* it is used.
- Every source file is readable by clicking its name anywhere on the page.
- Automated test suites cover the decomposition, the network layer, the band analysis and the site structure itself.

Computed live from the pipeline; nothing transcribed



(a) all 77 spectra are proper inverted parabolas, $q = 0$ at the apex (b) a tight band, $\Delta\alpha = 1.11$–1.25 (c–e) the apparent pore correlation (-0.50) collapses to $+0.10$ controlling for graph size; size survives at $+0.62$.

4. Technical Enrichment Activity Choices

Professional Online Certification Details (Individual Mandatory: 10–20 Hours)

S.No	Student Name	Chosen MOOC / Course Title	Platform	Hours
1	P. M. Radha Krishna	Algorithms on Graphs (BFS, shortest paths, connected components)	Coursera (UC San Diego)	20 Hrs
2	P. Kshitij Varma	Introduction to Complex Network Analysis	NPTEL	20 Hrs
3	G. Rohith Abhinav	Chemoinformatics / Computational Materials Science	NPTEL	16 Hrs
4	J. Keerthi Sree	Data Visualization with Python	Coursera (IBM)	15 Hrs
5	Isha	Machine Learning for Materials Informatics	edX / Coursera	15 Hrs

Rationale for these selections. Each maps to a component this project actually uses: **network analysis** (the graph representation and centrality measures), **complex networks** (fractal and multifractal characterisation), **chemoinformatics** (CIF handling, bond perception, MOF structure), **data visualisation** (the interactive platform), and **materials informatics** (descriptor-based screening — the project's stated goal).

Requirement: each student completes one approved 10–20 hour certification from a recognised platform relevant to the project.

5. Remaining Work & Timeline

Concrete engineering tasks, not open research questions

Priority 1 — Remove the size confound

- Normalise the fit window per structure so $\Delta\alpha$ stops growing with graph size; **or** compare only frameworks at matched graph size.
- Re-test the pore-diameter correlation afterwards. *This decides whether the band carries chemical meaning.*

Priority 2 — A genuinely diverse dataset

- Re-query CoRE MOF 2019 [9] with per-record database verification.
- Current set is 77 hMOF entries [8], **all Zn_4O** — hypothetical and single-metal, so no cross-metal claim is possible.

Priority 3 — Validate against measured properties

- Correlate the descriptor with published adsorption data to test whether it predicts anything, rather than merely describing structure.

Position of the work

The **machinery is complete and verified**: the decomposition, the periodic graph, the centrality analysis and the spectrum all work, are cross-validated across two implementations, and reproduce published crystallography.

What remains is establishing **whether the descriptor means anything physically** — and we have identified precisely what blocks that and how to remove it.

On reporting negative results

We report the size confound rather than the raw -0.497 correlation. A band built on an uncontrolled size effect would not survive scrutiny; identifying and quantifying it is the more defensible scientific position.

References (1 of 2)

Every entry is a source the project actually uses

[1] B. J. Bucior *et al.*, "Identification Schemes for Metal–Organic Frameworks to Enable Rapid Search and Cheminformatics Analysis," *Cryst. Growth Des.*, 19(11), 6682–6697, 2019. the decomposition we visualise

[2] J. Xiao, B. Wang, X. Li, "Deciphering the Generating Rules and Functionalities of Complex Networks," *Sci. Rep.*, 11, 22964, 2021. the method we implement

[3] C. Song, S. Havlin, H. A. Makse, "Self-Similarity of Complex Networks," *Nature*, 433, 392–395, 2005.

[4] O. Delgado-Friedrichs, M. O'Keeffe, "Identification of and Symmetry Computation for Crystal Nets," *Acta Cryst. A*, 59(4), 351–360, 2003. (Systre)

[5] M. O'Keeffe *et al.*, "The Reticular Chemistry Structure Resource (RCSR)," *Acc. Chem. Res.*, 41(12), 1782–1789, 2008. (net symbols)

[6] M. Eddaoudi *et al.*, "Modular Chemistry: Secondary Building Units...," *Acc. Chem. Res.*, 34(4), 319–330, 2001. (the SBU concept)

[7] N. S. Bobbitt, J. Chen, R. Q. Snurr, "MOFX-DB," *J. Chem. Eng. Data*, 68(2), 483–498, 2023. our data source

[8] C. E. Wilmer *et al.*, "Large-Scale Screening of Hypothetical MOFs," *Nature Chem.*, 4(2), 83–89, 2012. our 77 are hMOF = hypothetical

[9] Y. G. Chung *et al.*, "CoRE MOF 2019," *J. Chem. Eng. Data*, 64(12), 5985–5998, 2019. the set we still need

[10] B. Cordero *et al.*, "Covalent Radii Revisited," *Dalton Trans.*, 2832–2838, 2008. (bond-perception cutoffs)

References (2 of 2)

S. S.-Y. Chui *et al.*, *Science*, 283, 1148–1150, 1999. (HKUST-1 — test structure)

H. Li, M. Eddaoudi, M. O'Keeffe, O. M. Yaghi, *Nature*, 402, 276–279, 1999. (MOF-5 / IRMOF-1)

K. S. Park *et al.*, *PNAS*, 103(27), 10186–10191, 2006. (ZIF-8 — our chemical control)

J. H. Cavka *et al.*, *JACS*, 130(42), 13850–13851, 2008. UiO-66 — the 12-connected test

G. C. Shearer *et al.*, "Defect Engineering: Tuning the Porosity and Composition of UiO-66," *Chem. Mater.*, 28(11), 3749–3761, 2016. defects are real chemistry

M. Fiedler, "Algebraic Connectivity of Graphs," *Czech. Math. J.*, 23(2), 298–305, 1973. (λ_2)

U. Brandes, "A Faster Algorithm for Betweenness Centrality," *J. Math. Sociol.*, 25(2), 163–177, 2001.

A. H. Larsen *et al.*, "The Atomic Simulation Environment," *J. Phys. Condens. Matter*, 29(27), 273002, 2017. (ASE)

A. A. Hagberg, D. A. Schult, P. J. Swart, "Exploring Network Structure. . . using NetworkX," *SciPy2008*, 11–15. (NetworkX)

Full annotated list — what each reference is *for* — in `6_references/README.md`.

Thank You

Interactive platform, source code and full documentation:

github.com/Radha-Krishna-19/fyp